SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario Based Questions

Course Code:	DSA0405	**Course Title:**	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Scenario Based Questions	**Assessment:**	Assessment Tool 1– Scenario Based Questions
Weightage:	40% of CIA	**Total Marks:**	100
CO Statement:	Apply statistical estimation and hypothesis-testing techniques to solve real-world scenario-based problems involving confidence intervals and inferential decision-making. (BL4)		
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Section / Batch:	AIDS	**Date:**	18-08-2026

Code of Conduct

I VISHAAL M S / 192424420 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________________

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Problem 1 — Online Learning: Course Completion Time

An online learning platform measures the completion time of a course for 100 students. The mean completion time is 14 days and the standard deviation is 3 days. Construct a 95% confidence interval for the true mean completion time.

1. Population and Sample

Population: all students who take this course on the platform. Sample: the 100 students whose completion time was measured.

2. Parameter of Interest

The population mean course completion time, μ (in days).

3. Point Estimate

$$\bar{x} = 14 \text{ days}$$

4–5. Hypotheses

This is a pure estimation problem — no specific claimed or target completion time is given to test against, so no meaningful H_0 / H_1 applies here. (If the platform had advertised a target, e.g. "course completes in 12 days on average," the same interval could then be used to test H_0: $\mu = 12$ vs H_1: $\mu \neq 12$.)

6. Significance Level

$\alpha = 0.05$ (for a 95% confidence interval).

7. Confidence Interval

$$CI = \bar{x} \pm z_{0.025} \times (s / \sqrt{n}) = 14 \pm 1.96 \times (3 / \sqrt{100}) = 14 \pm 0.588$$

95% Confidence Interval: (13.41 days, 14.59 days)

8–9. p-value and Comparison with α

Not applicable — with no hypothesized benchmark value, there is no test statistic or p-value to compute here (see step 4–5).

10. Industry-Oriented Conclusion

The platform can be 95% confident that the true average course completion time lies between 13.41 and 14.59 days. This narrow range gives course designers and student-success teams a reliable benchmark for setting pacing expectations, scheduling reminder emails, and identifying students who are falling significantly behind the norm.

Problem 2 — Hospital: Patient Satisfaction

A hospital surveys 400 randomly selected patients, and 340 report that they are satisfied with the service. Construct a 95% confidence interval for the proportion of satisfied patients.

1. Population and Sample

Population: all patients treated by the hospital. Sample: the 400 randomly selected patients surveyed.

2. Parameter of Interest

The population proportion of patients who are satisfied with the service, p.

3. Point Estimate

$$\hat{p} = x / n = 340 / 400 = 0.85 \ (85\%)$$

4–5. Hypotheses

No target satisfaction rate is specified to test against, so this remains a pure estimation problem — no H_0 / H_1 applies. (A claim such as "patient satisfaction is 90%" could be tested against this same interval if provided.)

6. Significance Level

$\alpha = 0.05$ (for a 95% confidence interval).

7. Confidence Interval

$$CI = \hat{p} \pm z_{0.025} \times \sqrt{[\hat{p}(1-\hat{p})/n]} = 0.85 \pm 1.96 \times 0.0179 = 0.85 \pm 0.0350$$

95% Confidence Interval: (81.50%, 88.50%)

8–9. p-value and Comparison with α

Not applicable — no hypothesized benchmark satisfaction rate was provided (see step 4–5).

10. Industry-Oriented Conclusion

The hospital can be 95% confident that the true proportion of satisfied patients lies between 81.5% and 88.5%. This is a strong satisfaction range overall; hospital administration can use this as a baseline when evaluating the impact of future service improvements or when benchmarking against other healthcare providers.

Problem 3 — Telecom: Monthly Data Usage

A telecom company wants to estimate the average monthly data usage of its customers. A sample of 225 customers has a mean usage of 16 GB and a standard deviation of 5 GB. Construct a 99% confidence interval and interpret the result.

1. Population and Sample

Population: all customers of the telecom company. Sample: the 225 customers whose monthly data usage was measured.

2. Parameter of Interest

The population mean monthly data usage, μ (in GB).

3. Point Estimate

$$\bar{x} = 16 \text{ GB}$$

4–5. Hypotheses

No specific claimed average usage is given to test against, so this remains a pure estimation problem — no H_0 / H_1 applies. (A claim such as “average usage is 15 GB” could be tested against this same interval if provided.)

6. Significance Level

$\alpha = 0.01$ (for a 99% confidence interval).

7. Confidence Interval

$$CI = \bar{x} \pm z_{0.005} \times (s / \sqrt{n}) = 16 \pm 2.576 \times (5 / \sqrt{225}) = 16 \pm 0.859$$

99% Confidence Interval: (15.14 GB, 16.86 GB)

8–9. p-value and Comparison with α

Not applicable — no hypothesized usage benchmark was provided (see step 4–5).

10. Industry-Oriented Conclusion

The telecom company can be 99% confident that the true average monthly data usage across its customer base lies between 15.14 GB and 16.86 GB. This high-confidence estimate is well-suited for infrastructure and capacity-planning decisions, where the cost of underestimating demand is high; it can also guide the design of data-plan tiers that match typical customer usage.

Problem 4 — Customer Support: Complaint Resolution Time

A company claims that the average time taken by its customer-support team to resolve a complaint is 24 hours. A sample produces a 95% confidence interval of (25.2, 27.8) hours. Based only on this confidence interval, would you reject the company's claim? Explain.

1. Population and Sample

Population: all customer complaints handled by the support team. Sample: the complaints included in the study used to construct the given interval.

2. Parameter of Interest

The population mean complaint-resolution time, μ (in hours).

3. Point Estimate

Not given directly, but it can be inferred as the midpoint of the confidence interval:

$$\bar{x} \approx (25.2 + 27.8) / 2 = 26.5 \text{ hours} \quad (\text{margin of error} \approx \pm 1.3 \text{ hours})$$

4. Null Hypothesis (H_0)

H_0: $\mu = 24$ hours (the company's claim is accurate).

5. Alternative Hypothesis (H_1)

H_1: $\mu \neq 24$ hours (the true average resolution time differs from the claimed 24 hours).

6. Significance Level

$\alpha = 0.05$ (consistent with the given 95% confidence interval).

7. Confidence Interval / Decision Rule

The 95% confidence interval is given directly: (25.2, 27.8) hours. Using the CI-based decision rule: if the hypothesized value μ_0 falls inside the interval, we fail to reject H_0; if it falls outside, we reject H_0.

Is $\mu_0 = 24$ inside (25.2, 27.8)? — No, 24 < 25.2, so it falls outside the interval.

8–9. p-value and Comparison with α

The exact p-value cannot be computed without the raw sample size and standard deviation, but it can be reasoned directly from the interval: since the claimed value (24 hours) lies completely outside the 95% confidence interval, the p-value for testing $H_0: \mu = 24$ must be less than $\alpha = 0.05$.

10. Industry-Oriented Conclusion

Yes — based on this confidence interval alone, the company's claim should be rejected. Since the entire interval (25.2 to 27.8 hours) lies above 24 hours, the data provide strong evidence that the true average resolution time is longer than advertised. From a business standpoint, the company is likely understating its resolution time to customers; it should either correct its published SLA or investigate and address the operational delays causing the gap.

Problem 5 — Manufacturing: Packaged Product Weight

A manufacturer claims that the average weight of a packaged product is 500 grams. A 95% confidence interval based on a sample is (496, 502) grams. Is the claimed value of 500 grams consistent with the sample evidence? Explain your answer.

1. Population and Sample

Population: all packaged units of the product produced by the manufacturer. Sample: the units included in the study used to construct the given interval.

2. Parameter of Interest

The population mean package weight, μ (in grams).

3. Point Estimate

Not given directly, but it can be inferred as the midpoint of the confidence interval:

$$\bar{x} \approx (496 + 502) / 2 = 499 \text{ grams} \quad (\text{margin of error} \approx \pm 3 \text{ grams})$$

4. Null Hypothesis (H_0)

$H_0: \mu = 500$ grams (the manufacturer's claim is accurate).

5. Alternative Hypothesis (H_1)

$H_1: \mu \neq 500$ grams (the true average package weight differs from the claimed 500 grams).

6. Significance Level

$\alpha = 0.05$ (consistent with the given 95% confidence interval).

7. Confidence Interval / Decision Rule

The 95% confidence interval is given directly: (496, 502) grams. Using the CI-based decision rule: if the hypothesized value μ_0 falls inside the interval, we fail to reject H_0; if it falls outside, we reject H_0.

Is $\mu_0 = 500$ inside (496, 502)? — Yes, 496 < 500 < 502, so it falls inside the interval.

8–9. p-value and Comparison with α

The exact p-value cannot be computed without the raw sample size and standard deviation, but since the claimed value (500 grams) lies well within the 95% confidence interval — close to its midpoint — the p-value for testing $H_0: \mu = 500$ must be greater than $\alpha = 0.05$.

10. Industry-Oriented Conclusion

Yes — the claimed value of 500 grams is consistent with the sample evidence. Since 500 falls comfortably inside the 95% confidence interval (496, 502), and is in fact very close to its midpoint (499 grams), there is no statistical evidence to suggest the true average package weight differs from the manufacturer's claim. From a quality-control standpoint, the packaging process appears to be operating in line with its stated specification, and no corrective action is warranted based on this result.